

Lorenzo Remy- Mechanical Systems Design, Build & Test Portfolio

Preface

This portfolio presents a selection of build-focused work across aerospace development, industrial plant maintenance, and mechanical system assembly, reflecting my ability to contribute at multiple stages of a hardware lifecycle. Together, these experiences shape a systems-oriented approach that connects analytical design to manufacturability, assembly, validation, and sustainment, allowing me to apply mechanical engineering methods whether I am responsible for the full cradle-to-grave execution or for a critical phase within a larger system.

1. NASA Student Launch Initiative (Loyola Marymount University)

As Mechanical Design Lead for LMU's NASA Student Launch capstone project, I led a 6-person team in designing, building, and flight-testing a sub-scale and full-scale high-powered rocket.

Role & Responsibilities:

- Design & Manufacture: Led CAD design (SolidWorks) and 3D printing of critical components including the avionics bay/tray, bulkheads, motor mount, fins, and subsystem alignment hardware.
- Subsystem Integration: Packaged and interfaced recovery system (main/drogue parachutes with ejection charges), avionics (PCBA with power units), propulsion (L1 & L2 solid propellant motors), and airbrake subsystem.
- Testing & Validation: Created subsystem test plans and redundancy tests per NASA specifications; developed launch plans to validate full vehicle ecosystem performance.

Key Technical Achievements:

- Reduced vehicle mass by 25% through iterative design optimization and material selection (PLA → PETG after subscale testing proved similar thickness of PLA insufficient for landing impact)
- Designed custom rail-like alignment system for center body tube, ensuring avionics tray sat flush with airbrake system while limiting torsional forces from servo motor actuation.
- Redesigned bulkheads from integrated to modular components after subscale testing, achieving more rigid printed products and improved assembly reliability.

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Figure 1 (Top): Sub-scale Rocket (L1 Solid Propellant Motor) (3D Printed Vehicle)

Figure 2 (Bottom): Full-scale Rocket (L2 Solid Propellant Motor) (3D Printed Vehicle)

2. Boeing 777-8F Main Deck Cargo Door Safety Strap – Full-Scale Mockup

As a Systems Engineering Intern at Boeing's Everett Factory, my team was tasked with the design and construction of a scaled mockup for a load-bearing safety restraint system for the 777-8F Main Deck Cargo Door (MDCD).

Role & Responsibilities:

- Sourced internal airplane dimensions for the appropriate airframe section to guide mockup specifications.
- Identified potential mounting points on existing airframe while considering door interface impacts and internal cargo capacity constraints.
- Led hands-on construction: coordinated woodwork and foam cutting with shop, selected fasteners, and assembled MDCD area with a 6-person team.
- Tested multiple iterations with various quick-connect/release hardware configurations; conducted load testing to validate customer specifications.

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3. Siemens Centrifugal Compressor Overhaul – Industrial Process Plant

At Reliability Maintenance Services Limited, I worked on teams performing complete overhauls of industrial-scale rotating equipment, including Siemens centrifugal compressors.

Role & Responsibilities:

- Performed alignment and leveling of shafts to precise tolerances under guidance of senior inspectors.
- Replaced seals, lubricants, and rotating equipment (gears) while ensuring proper facing of nameplates and directional arrows.
- Worked in accordance with API 617 standards for centrifugal compressor maintenance.



Figure 3 (Left): Siemens compressor referenced in statement

Figure 4 (Right): Teammates working on compressor referenced in statement

4. Additional Mechanical Systems Build/Assembly Experience

Rapid Oilfield Services Limited: Assembled, inspected, and maintained manual handling tools including drill pipe slips, elevators, tongs, and safety clamps. Applied storage/running lubricants and configured tools with appropriate dies and inserts.

Cooling Tower Fan Maintenance: As part of a 5-person team, visually inspected fan shrouds and blades. Removed and replaced damaged blades, aligning them to proper pitch angles for optimum operation.